

IPE 4111
Industrial Management

Technology management

What is Technology Management?



- Technology refers to Information Technology (IT), which includes computer hardware/software, telecommunications, networks, and database.
- **Technology Management** can also be defined as the integrated planning, design, optimization, operation and control of technological products, processes and services, a better definition would be the management of the use of technology for human advantage.

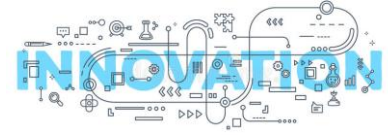


Key Tasks of MoT

The key tasks of MoT or Management of Technology are:

- Technology Planning like deciding technology strategy, selecting appropriate technologies etc.
- R&D Management
- Innovation Management
- Strategic Management of Technology (SMoT)

Innovation



- **Invention** can be defined as the creation of a product or introduction of a process for the first time.
- **Innovation** occurs if someone improves on or makes a significant contribution to an existing product, process or service.
- Steve Jobs' iPod is cited as an example of innovation at its best. The iPod wasn't the first portable music device (Sony popularized the “music anywhere, anytime” concept 22 years earlier with the Walkman); the iPod wasn't the first device that put hundreds of songs in our pocket.
- What made Apple innovative was that it combined all of these elements — **design, ergonomics** and **ease of use** — in a single device, and then tied it directly into a platform that effortlessly kept that device updated with music.

Innovation Management



Venkatakrishnan Balasubramanian, a research analyst with Infosys Labs has offered five considerations to innovation management:

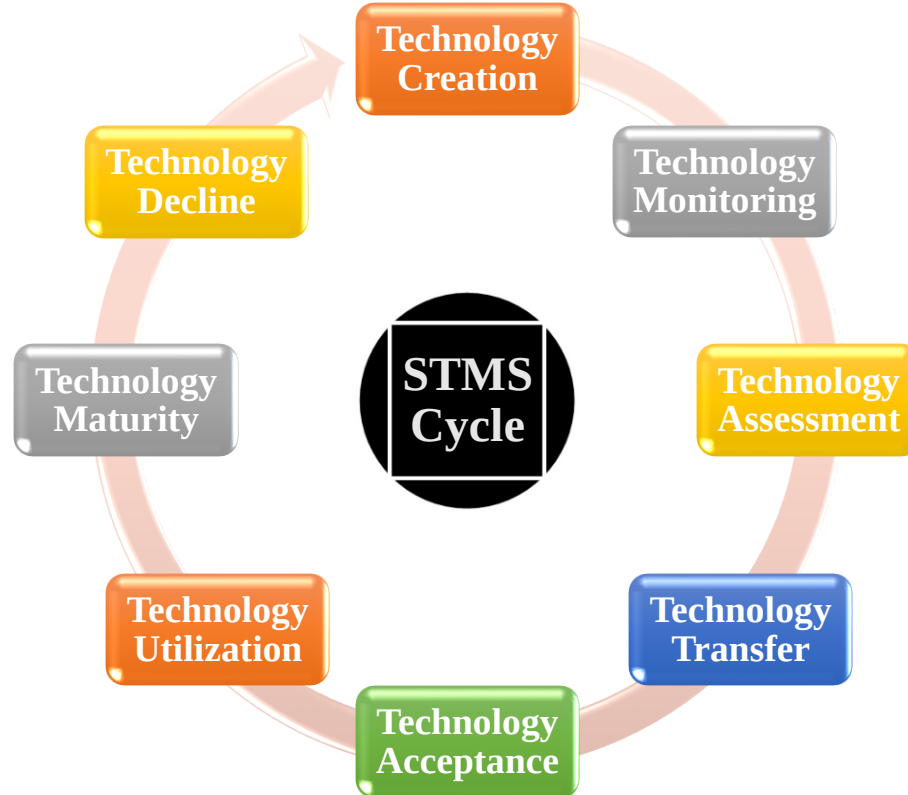
1. **Competitive Advantage** over other industries.
2. **Business Alignment** or the differentiating factors of your innovation should be conceptualized around the key strategic focus of the enterprise and its goals.
3. Knowing the **Customers** who will benefit from your innovation is paramount.
4. Identifying resources, processes, risks, partners and suppliers and **Execution** of these in the market are also important for succeeding in the innovation.
5. **Business Value** or assessing the value (monetary, market size, etc.) of the innovation and how the idea will bring that value into the organization is a critical underlying factor in selecting which idea

Strategic Technology Management System

- Strategic Technology Management System (STMS) emphasizes systematic management of technology (SMoT) on long term basis as the technology moves along its life cycle from birth to decline. It includes eight phases.



Phases of STMS



Phases of STMS



- **Technology Creation:** This phase involves creation & generation of new technologies. This phase involves creativity, invention, innovation and developing requisite & supportive corporate culture for promoting technology creation and generation.
- **Technology Monitoring:** This phase calls for monitoring technology trends and changes before implementing new technology. This phase involves Installing & developing information systems for monitoring technological trends and changes, Competitive analysis and Customer & supplier interfaces to understand market and technological changes.

Technology Transfer: This phase leads to transfer of technology from external source to own Research and Development (R&D); and internally from R&D to production. This phase involves Entering strategic alliances to develop or acquire potential technologies, Using product design teams effectively, Reducing functional barriers to technology transfer, etc.

Phases of STMS



- **Technology Acceptance:** This phase calls for acceptance of technology as a beneficial change and involves Supportive organizational design & structures, Senior management commitment, Supportive corporate culture, etc.
- **Technology Utilization:** This phase leads to technology growth as reflected by increase in sales. This phase involves Effective project management for seeking maximum utilization, Supportive marketing strategies, efforts and utilizing feedback for improvement, etc.

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Technology Life Cycle (TLC)



- The **Technology Life Cycle (TLC)** describes the costs and profits of a product from technological development phase to market maturity to eventual decline.
- Due to rapidly increasing rates of innovation, products such as electronics and pharmaceuticals in particular are vulnerable to shorter life cycles. Thus TLC is focused primarily on the time and cost of development as it relates to the projected profits. TLC can be described as having four distinct stages.

Technology Life Cycle (TLC)

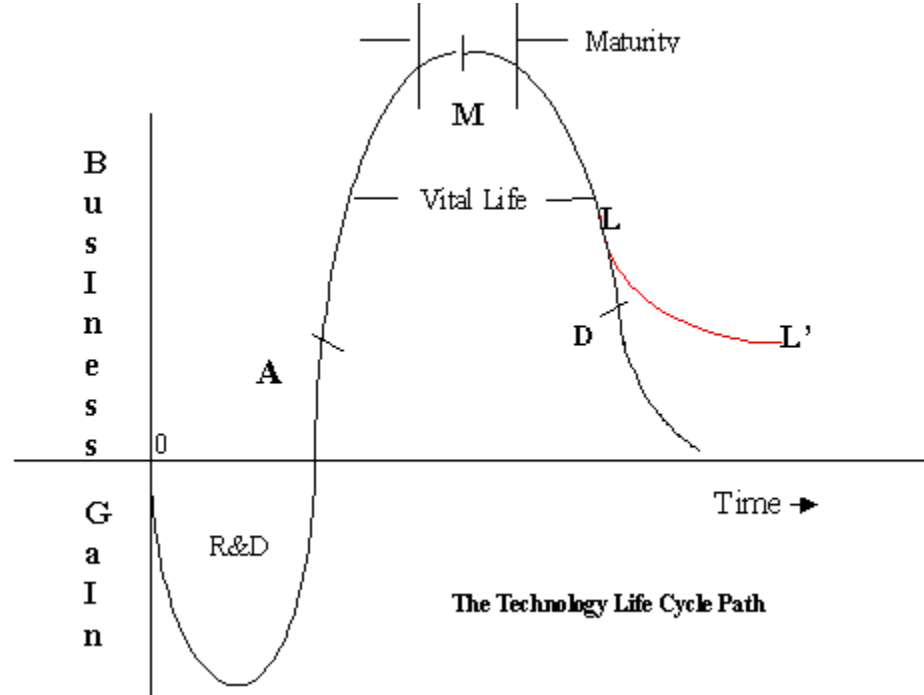


Illustration of the stages in the technological life cycle

Four Stages of TLC



- **Research and Development** – During this stage, risks are taken to invest in technological innovations. By strategically directing R&D towards the most promising projects, companies and research institutions slowly work their way toward beta versions of new technologies.
- **Ascent Phase** – This phase covers the timeframe from product invention to the point at which out-of-pocket costs are fully recovered. At this junction the goal is to see to the rapid growth and distribution of the invention and leverage the competitive advantage of having the newest and most effective product.

Four Stages of TLC

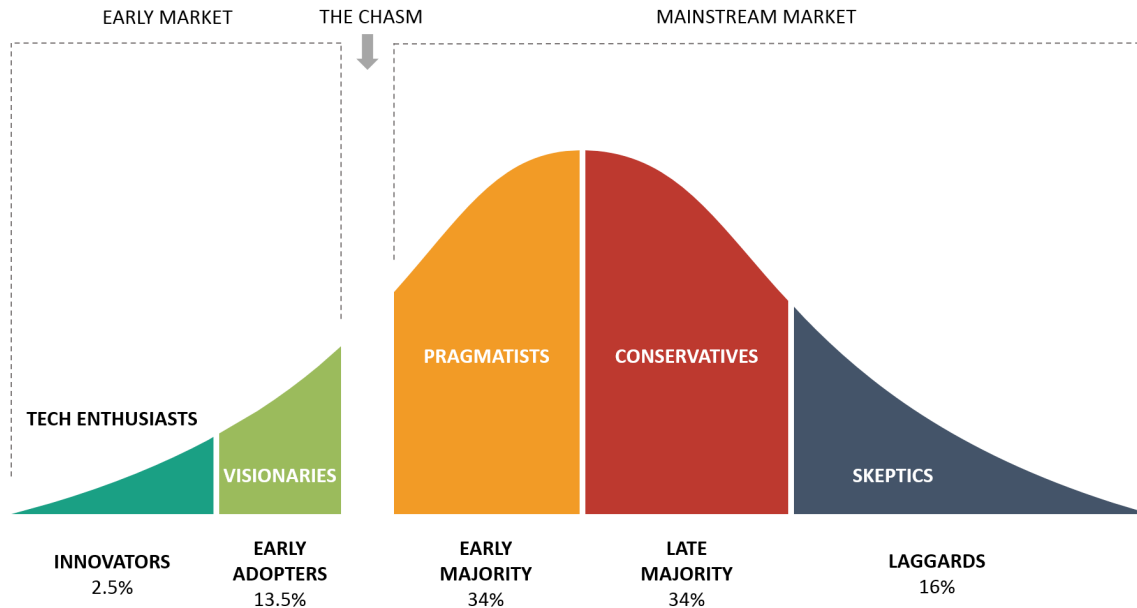


- **Maturity Stage** – As the new innovation becomes accepted by the general population and competitors enter the market, supply begins to outstrip demand. During this stage, returns begin to slow as the concept becomes normalized.
- **Decline (or Decay) Phase** – The final phase is when the utility and potential value to be captured in producing and selling the product begins dipping. This decline eventually reaches the point of a zero-sum game, where margins are no longer procured.

Technology Adoption Life Cycle



- **Technology Adoption Life Cycle** highlights the way in which consumers embrace new products and services.



Technology Adoption Life Cycle



- **Innovators** – These are risk-oriented, leading-edge minded individuals who are extremely interested in technological developments (often within a particular industry). Innovators are a fractional segment of the overall consumer population.
- **Early Adopters** – A larger but still relatively small demographic, these individuals are generally risk-oriented and highly adaptable to new technology. Early adopters follow the innovators in embracing new products, and tend to be young and well-educated.
- **The chasm** refers to the technology adoption lifecycle, or the transition from the early market into the mainstream sight. Crossing the chasm means the opportunity for hyper-growth and market success. It's the leap from being a new, little-known and exploratory product, to mass adoption and well-known status.

Technology Adoption Life Cycle



- **Early Majority** – Much larger and more careful than the previous two groups, the early majority are open to new ideas but generally wait to see how they are received before investing.
- **Late Majority** – Slightly conservative and risk-averse, the late majority is a large group of potential customers who need convincing before investing in something new.
- **Laggards** – Extremely frugal, conservative, and often technology-averse, laggards are a small population of usually older and uneducated individuals who avoid risks and only invest in new ideas once they are extremely well-established.

Case Studies



- Companies that experience innovation success grab onto it and believe that it is their secret to everlasting success. Unfortunately, this is not the case, although it would definitely make a CEOs job a lot easier.
- There are some big companies, which is the victim of technology failure.
- Some examples of famous corporations that suffered from innovation failure, including companies that have gone out of business will be discussed here.
- **For self-study:**
<https://www.valuer.ai/blog/50-examples-of-corporations-that-failed-to-innovate-and-missed-their-chance>

Xerox



- Xerox was actually first to invent the PC and their product was way ahead of its time. Unfortunately, the management thought going digital would be too expensive and they never bothered to exploit the opportunities they had.
- The CEO David Kearns was convinced that the future of Xerox was in copy machines. The digital communication products invented weren't seen as something that could replace black marks on white paper. Xerox failed to understand that you can't keep perpetually making money on the same technology. Sometimes technology fails too.

Kodak



- In 1975, a young engineer at Kodak named Steve Sasson invented the digital camera. At first, it wasn't much to look at, weighing eight pounds and able to produce pictures of only 0.01 megapixels (today's iPhones produce 12 megapixels images, a 120,000% increase). He estimated at the time that it would take 15 to 20 years for the technology to become viable.
- Being a silly big company with its head in the sand, the firm failed to pursue digital photography. New upstarts took over the market and prospered, while Kodak vanished into irrelevance. The company filed for bankruptcy in 2011.

Kodak



- Once again, the truth is more complex. The company did, in fact, pursue the digital photography business in a serious way. It also made big investments in quality printing for digital photos. The problem was that it made most of its money on developing film, a business that completely disappeared.
- It's tough to say what Kodak could have done. The only company that really seems to have profited from digital photography is Facebook and it's hard to see how Kodak would have had a competitive advantage in building social networks. The only real solution would have been to invent an entirely new business that could replace its *cash cow* (investment).

Nokia



- Nokia, a company founded in Finland was the first to create a cellular network in the world. In the late 1990s and early 2000s, Nokia was the global leader in mobile phones.
- With the arrival of the Internet, other mobile companies started understanding how data, not voice, was the future of communication. Nokia didn't grasp the concept of software and kept focusing on hardware because the management feared to alienate current users if they changed too much.

Nokia



- Nokia's mistake was the fact that they didn't want to lead the drastic change in user experience. This caused Nokia to develop a mess of an operating system with a bad user experience that just wasn't a fit on the market.
- The company overestimated the strength of its brand and believed they could arrive late in the smartphone game and succeed. In 2007 Steve Jobs launched the iPhone, a phone without a keyboard, which was revolutionary at the time. Really, watch the video and listen to people losing their minds the first time they watch someone using a touchscreen. In 2008 Nokia finally made the decision to compete with Android, but it was too late. Their products weren't competitive enough.

Blockbuster



- The video-rental company was at its peak in 2004. They survived the change from VHS to DVD but failed to innovate into a market that allowed for delivery (much less streaming).
- While Netflix was shipping out DVD's to their consumer's homes, Blockbuster figured their physical stores were enough to please their customers. Because they had been the leader of the movie rental market for years, management didn't see why they should change their strategy.

Blockbuster



- Back in 2000, the founder of Netflix Reed Hastings proposed a partnership to the former CEO of Blockbuster John Antioco. Netflix wanted Blockbuster to advertise their brand in the stores while Netflix would run Blockbuster online. The idea got turned down by Antioco because he thought it was ridiculous and that Netflix's business model was "niche business." Little did he know that Hasting's idea would have saved Blockbuster. In 2010 Blockbuster filed for bankruptcy and Netflix is now a \$28 billion dollar company.
- The Forbes article that aptly describes what exactly happened to Blockbuster, "**The internet didn't kill Blockbuster, the company did it to itself**"

Borders

BORDERS

Beyond Books

- Borders Group opened its first bookstore in 1971 and they were a success for years. But in the mid- 2000s Borders failed to adapt to new technologies and never embraced the Internet like Amazon and Barnes & Noble. As more people began to order their books online, the fading of a giant like Borders was inevitable.
- Amazon Kindle came out on in 2007, Barnes & Noble debuted Nook in 2009 and Apple's iPad launched in 2010. Border's Kobo (which is still alive) came out a little too late in 2011. It wasn't just the Internet and competition that killed the company, it was their overall strategy. Borders opened too many stores making it harder to shed unprofitable locations and there were many. In 2011 Borders filed for bankruptcy, closing 399 stores and laying off 10,700 employees.

Thank
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